Task 5

Fill in the table to compare some of the different ways that electricity can be generated.

Energy source	Maximum power output	Start-up time	Reliability (when is it available?)	Suitable for charging a cell in an isolated road sign?
gas-fired power station	very large	fast	always available	
nuclear power station				no – too much power available
small-scale hydroelectric supply	small	very fast – just turn on a tap		
small bank of solar panels				
single wind turbine				

Discussion

- Is there one way to generate electricity that will meet all of our needs?
- If you had to plan a policy to meet our country's energy needs what would you suggest?

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Discussion

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Task 5

Fill in the table to compare some of the different ways that electricity can be generated. Decide what factors would be useful to compare and write the headings in the first row.

Energy source				
gas-fired power station				
nuclear power station				
small-scale hydroelectric supply				
small bank of solar panels				
single wind turbine				

Discussion

- Is there one way to generate electricity that will meet all of our needs?
- If you had to plan a policy to meet our country's energy needs what would you suggest?

Suggested answers and teaching notes

Choose the most appropriate table for Task 5, depending on the amount of challenge required by your students.
The following links would be useful to students researching the advantages and disadvantages of the main energy resources;

- www.bbc.co.uk/education/guides/zpp4jxs/revision/1
- www.bbc.co.uk/education/guides/znn9q6f/revision
- Using solar and wind energy at home - BBC clip www.bbc.co.uk/education/clips/zwfqmp3

Energy source	Maximum power output	Start-up time	Reliability (when is it available?)	Suitable for charging a cell in an isolated road sign?
gas-fired power station	very large (up to 4 000 MW)	fast	always available	no
nuclear power station	very large (500 – 4 000 MW)	slow	always available	no – too much power available
small-scale hydroelectric supply	small (up to 10 MW)	very fast – just turn on a tap	depends on type and water availability	no
small bank of solar panels	small (up to 4 000 kW)	fast	reliant on sunlight	yes
single wind turbine	small (up to 8 MW)	fast	dependent on wind	no